TV Advertising vs Sales – Linear Regression Report

This report analyses the relationship between TV advertising budget and sales using simple linear regression and formal statistical inference.

First Five Rows of the Dataset

TV	sales
230.1	22.1
44.5	10.4
17.2	9.3
151.5	18.5
180.8	12.9

Descriptive Statistics
TV Mean: 147.0425, Median: 149.7500, Mode: 17.2000, SD: 85.8542
Sales Mean: 14.0225, Median: 12.9000, Mode: 9.7000, SD: 5.2175

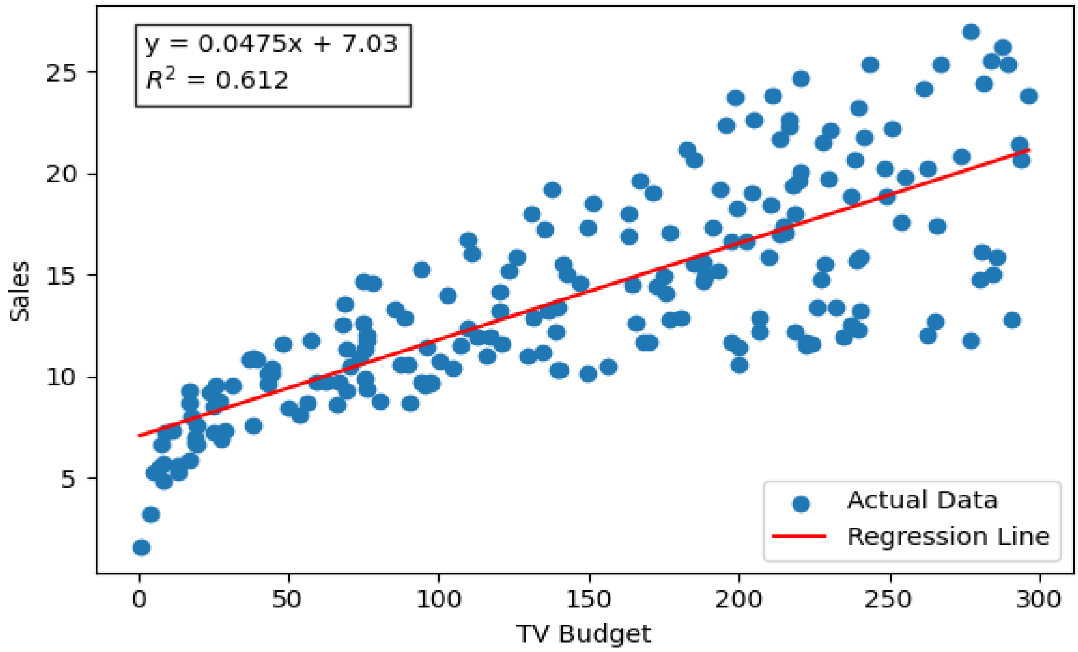
Regression Results
Slope = 0.047537
Intercept = 7.032594
RSS = 2102.5306
RSE = 3.2587
R = 0.7822
R^2 = 0.6119

T■Tests (Two■Tailed)
Slope: $t = 17.6676$, $p < 0.001$
Intercept: $t = 15.3603$, $p < 0.001$

95% Confidence Interval for Slope
[0.042231, 0.052843]

Interpretation
You can be 95% confident that for every $1000 spent on TV advertising, sales increase between 42 and 53 units.

Regression Plot



y = 0.0475x + 7.03
R^2 = 0.612
Actual Data
Regression Line
Sales
TV Budget
25
20
15
10
5
0
50
100
150
200
250
300